

CHEM 231 — Pre-OCChem Readiness Exam

Gabe · gauges what you actually know before Organic Chemistry I. Blank-page format on purpose.

How to take it: No notes, no app, no lookups — this measures cold recall, which is the only kind that counts in a real exam. Work on paper (or annotate this PDF in GoodNotes). **Draw structures fully** — every bond and lone pair. Budget ~75 min but don't rush. When done, photograph your work and send it to Claude in the CHEM 231 chat: you'll get a scored, **prioritized gap report** (which topics to study first, tied to the app's Learn tab and your Klein 5e sections).

Scoring by section (self-check with the key, or let Claude do it): each section flags **green** ($\geq 80\%$ — solid), **yellow** ($50\text{--}79\%$ — review the Learn primer), **red** ($< 50\%$ — study first, before anything else).

Section 1 · Valence electrons & Lewis structures foundation

1. State the number of valence electrons for each: H, C, N, O, F, Cl.

6 pts

2. Draw the complete Lewis structure of formaldehyde, CH_2O . Show every bond and lone pair, and state the total valence electrons you used.

5 pts

3. Draw the Lewis structure of the cyanide ion, CN^- . Show lone pairs and where the charge sits.

5 pts

4. Draw methanol, CH_3OH — all atoms, bonds, and lone pairs.

4 pts

5. For CO_2 : draw the Lewis structure, state the total valence electrons, and explain in one line what you did when the central carbon came up short after drawing single bonds.

5 pts

Section 2 · Formal charge your known weak spot

6. Compute the formal charge on O in hydroxide, OH^- . Show the three inputs: valence, lone-pair electrons, bonds.

4 pts

7. Compute the formal charge on N in ammonium, NH_4^+ . Show your inputs.

4 pts

8. Compute the formal charge on O in hydronium, H_3O^+ . Show your inputs.

4 pts

9. (a) A carbon with 3 bonds and 1 lone pair — formal charge, and what is this species called? (b) A carbon with 3 bonds and 0 lone pairs — formal charge, and name?

6 pts

Section 3 · Polarity & molecular dipoles

10. Rank these by electronegativity, most to least: C, F, H, N, O.

4 pts

11. Is the $\text{C}=\text{O}$ bond polar? If so, which atom carries δ^- , and why?

3 pts

12. CO_2 has two very polar bonds. Is the *molecule* polar? Explain in one sentence using its geometry.

4 pts

13. Is H_2O a polar molecule? Explain why your answer differs from CO_2 even though both have two polar bonds.

4 pts

Section 4 • Hybridization & VSEPR

14. Give the hybridization and geometry (shape name + approx. bond angle) of the carbon in CH_4 . 3 pts

15. Same for a carbon in ethene, $\text{CH}_2=\text{CH}_2$. 3 pts

16. Same for a carbon in ethyne, $\text{HC}\equiv\text{CH}$. 3 pts

17. Give the hybridization of N in NH_3 , and the molecular *shape name* + angle. (Careful: the shape name is not "tetrahedral.") 4 pts

Section 5 • Intermolecular forces

18. Name the three intermolecular forces, weakest to strongest. 3 pts

19. What two structural features must a molecule have to hydrogen-bond *with itself*? 4 pts

20. Which boils higher: ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) or dimethyl ether (CH_3OCH_3)? They share the formula $\text{C}_2\text{H}_6\text{O}$. Explain. 4 pts

21. Which boils higher: butane or octane? Name the force responsible and why it differs. 3 pts

Section 6 • Integration & transfer

the real test

22. For methylamine, CH_3NH_2 : (a) draw the Lewis structure, (b) give the hybridization of C and of N, (c) state whether the molecule is polar, (d) name its strongest intermolecular force. 8 pts

23. Draw two different constitutional isomers with the formula $\text{C}_2\text{H}_6\text{O}$, and circle the one that can hydrogen-bond to itself. 6 pts

24. For the acetate ion, CH_3COO^- : draw a Lewis structure, assign the formal charge to the correct atom, and state (one sentence) whether the negative charge sits on one oxygen or is shared between both — and why. 7 pts

■ STOP ■

Answer key begins on the next page.

Do the whole exam first — the gap report is only accurate if you didn't peek.

Answer Key & Gap-Priority Guide

Check each section, tally your score, then read the priority guide at the bottom.

Section 1 — Valence & Lewis

1. $H = 1$, $C = 4$, $N = 5$, $O = 6$, $F = 7$, $Cl = 7$. (Valence = group number for main-group atoms.)
2. $H_2C=O$: carbon double-bonded to O, single-bonded to two H's; oxygen carries **2 lone pairs**. Total valence $e^- = 4 + 2(1) + 6 = 12$.
3. $[C\equiv N]^-$: triple bond between C and N; **one lone pair on C**, one lone pair on N. Total $e^- = 4 + 5 + 1$ (charge) = 10. Carbon carries the -1 (FC on C = $4 - 2 - 3 = -1$).
4. H_3C-O-H : carbon bonded to 3 H's and the O; oxygen bonded to C and H with **2 lone pairs**. Total $e^- = 4 + 4(1) + 6 = 14$.
5. $O=C=O$; total $e^- = 4 + 2(6) = 16$. After single bonds carbon is short of an octet, so you **convert a lone pair from each oxygen into a second bond**, giving two $C=O$ double bonds. Each O keeps 2 lone pairs; carbon reaches 8.

Section 2 — Formal charge

6. O in OH^- : 1 bond, 3 lone pairs. $FC = 6 - 6 - 1 = -1$.
7. N in NH_4^+ : 4 bonds, 0 lone pairs. $FC = 5 - 0 - 4 = +1$.
8. O in H_3O^+ : 3 bonds, 1 lone pair. $FC = 6 - 2 - 3 = +1$.
9. (a) $4 - 2 - 3 = -1$, a **carbanion**. (b) $4 - 0 - 3 = +1$, a **carbocation**. Same bond count, opposite charge — the lone pair decides. (3 bonds + 1 unpaired electron would be a neutral radical.)

Section 3 — Polarity & dipoles

10. $F > O > N > C \approx H$ (C and H are close; F most electronegative).
11. Yes, polar. O is δ^- (more electronegative, pulls the shared electrons); C is δ^+ .
12. **Nonpolar molecule**. CO_2 is linear, so the two equal, opposite bond dipoles cancel $\rightarrow$ net dipole zero.
13. **Polar**. Water is bent ($\sim 104.5^\circ$), so the two O-H bond dipoles don't cancel — they add to a net dipole. Geometry, not bond polarity, is the deciding difference from CO_2 .

Section 4 — Hybridization & VSEPR

14. CH_4 carbon: 4 domains $\rightarrow sp^3$, tetrahedral, 109.5° .
15. Ethene carbon: 3 domains (the double bond counts once) $\rightarrow sp^2$, trigonal planar, 120° .

16. Ethyne carbon: 2 domains → **sp**, linear, 180°.

17. N in NH₃: 4 domains (3 bonds + 1 lone pair) → **sp³**; molecular shape = **trigonal pyramidal**, ~107°. (The lone pair counts for hybridization but the shape names only the atoms.)

Section 5 — Intermolecular forces

18. London dispersion < dipole–dipole < hydrogen bonding.

19. (1) an H bonded to N, O, or F (donor), and (2) a lone pair on an N, O, or F (acceptor).

20. **Ethanol** (bp ≈ +78 °C) > dimethyl ether (≈ –24 °C). Ethanol has an O–H, so it hydrogen-bonds to itself; the ether's H's are all on carbon, so no self H-bonding — only dipole/dispersion.

21. **Octane** boils higher. Both are nonpolar (London dispersion only), but octane is larger with more surface area → stronger dispersion.

Section 6 — Integration

22. CH₃NH₂: C bonded to 3 H + N; N bonded to C + 2 H with 1 lone pair. (a) drawn as such; (b) C = sp³, N = sp³; (c) polar (the C–N and N–H bonds + lone pair give a net dipole); (d) strongest IMF = **hydrogen bonding** (N–H present).

23. The two C₂H₆O isomers: **ethanol** CH₃CH₂OH (has O–H → circle this one, self H-bonds) and **dimethyl ether** CH₃OCH₃ (no O–H).

24. Acetate CH₃COO[–]: the –1 sits on oxygen. The charge is **shared equally between both oxygens** by resonance (two equivalent structures) — each C–O bond is identical (bond order 1.5). That delocalization is why acetic acid's proton is acidic.

Your gap-priority guide

Section	If red (<50%)	Study first (in order)
2 · Formal charge	Highest priority — it underlies charges, mechanisms, everything downstream	Learn tab → Formal charge primer; run the app's FC Mastery Check to 2 clean days
1 · Valence & Lewis	Foundation — nothing else works without it	Learn tab → Valence, then Lewis; do the Lewis Lab molecules
4 · Hybridization/VSEPR	Domain-counting; the lone-pair trap	Learn tab → Hybridization; drill h-questions
3 · Polarity	Bond-vs-molecule distinction	Learn tab → Polarity; drill p-questions
5 · IMFs	Lightest; mostly rules	Learn tab → IMFs; drill i-questions
6 · Integration	If sections 1–5 are green but 6 is red, you know the pieces but not how to combine them — do more Lewis Lab + mixed drills	—

Rule of thumb: a red in Section 1 or 2 gets fixed before touching anything else — they're load-bearing. Yellow sections get one Learn-primer pass plus a drill round. Green sections need only occasional review to stay warm. Send your work to Claude for the exact ranked plan.